

1. System Architecture Diagram



Components:

Client – Web browser used by clients, technicians, and admins

Server – Handles requests, processes logic, and manages system operations

Database – Stores work orders, users, logs, and other system data

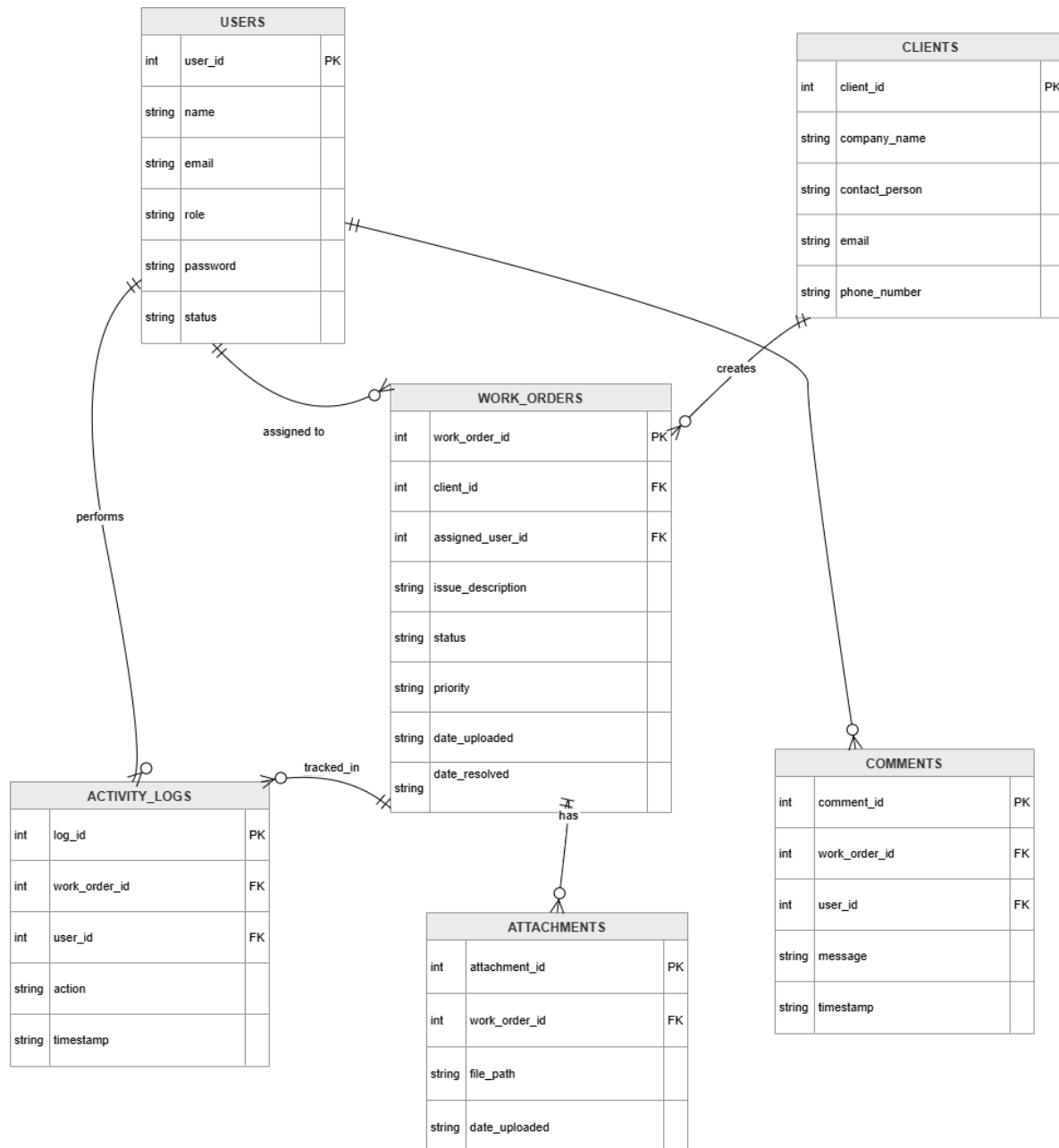
Network – Internet or local network connecting users to the system

Data Flow Explanation:

Users access the system through a web browser and submit requests such as creating or updating work orders. These requests are sent through the network to the application server, where they are processed. The server then interacts with the database to store or retrieve data. The processed information is sent back to the client and displayed to the user in real time.

2. Database Design

A. Entity Relationship Diagram



B. Database Tables

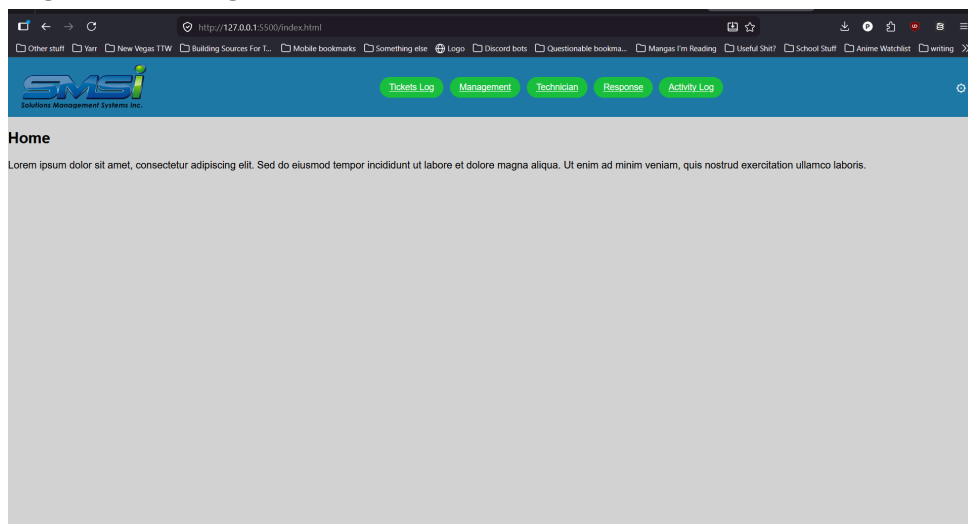
- Users
- Clients
- Work_Orders
- Activity_Logs
- Comments
- Attachments

C. Primary Keys and Relationships

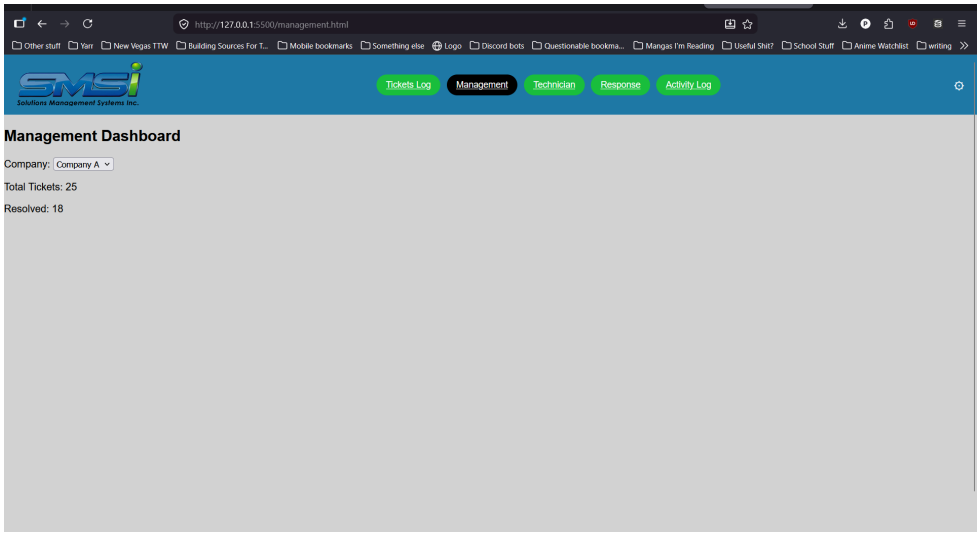
- **Users**
 - PK: user_id
- **Clients**
 - PK: client_id
- **Work_Orders**
 - PK: work_order_id
 - FK: client_id → Clients
 - FK: assigned_user_id → Users
- **Activity_Logs**
 - PK: log_id
 - FK: work_order_id → Work_Orders
 - FK: user_id → Users
- **Comments**
 - PK: comment_id
 - FK: work_order_id → Work_Orders
 - FK: user_id → Users
- **Attachments**
 - PK: attachment_id
 - FK: work_order_id → Work_Orders

3. System Interface Design

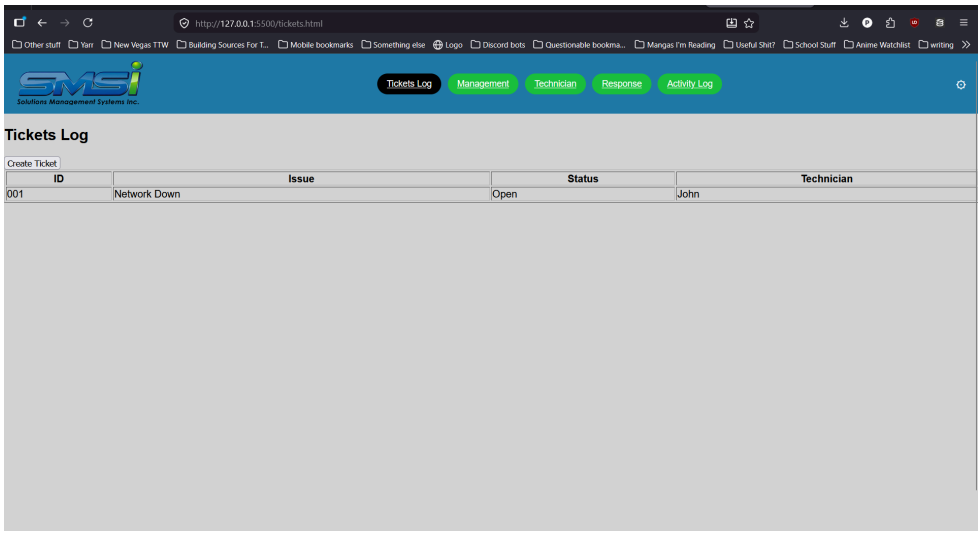
Login/ Home page:



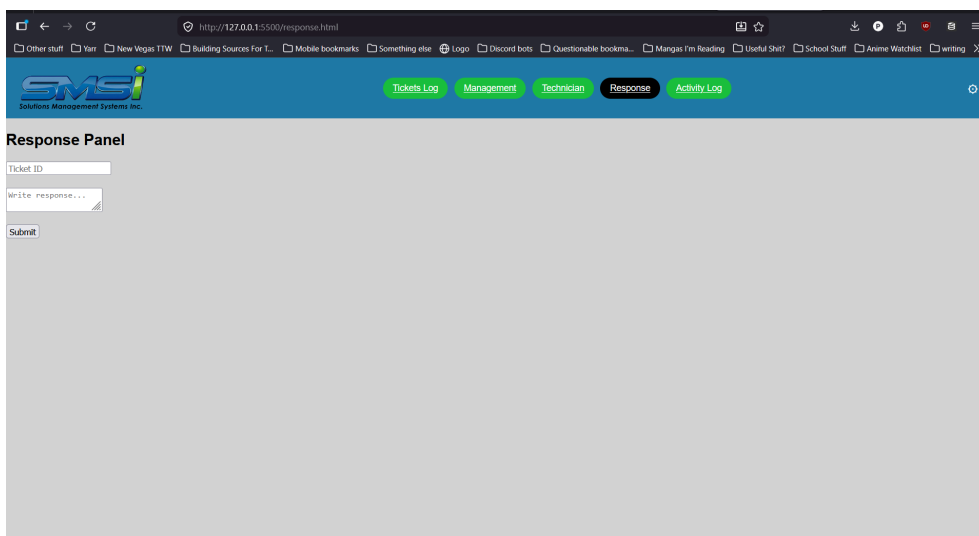
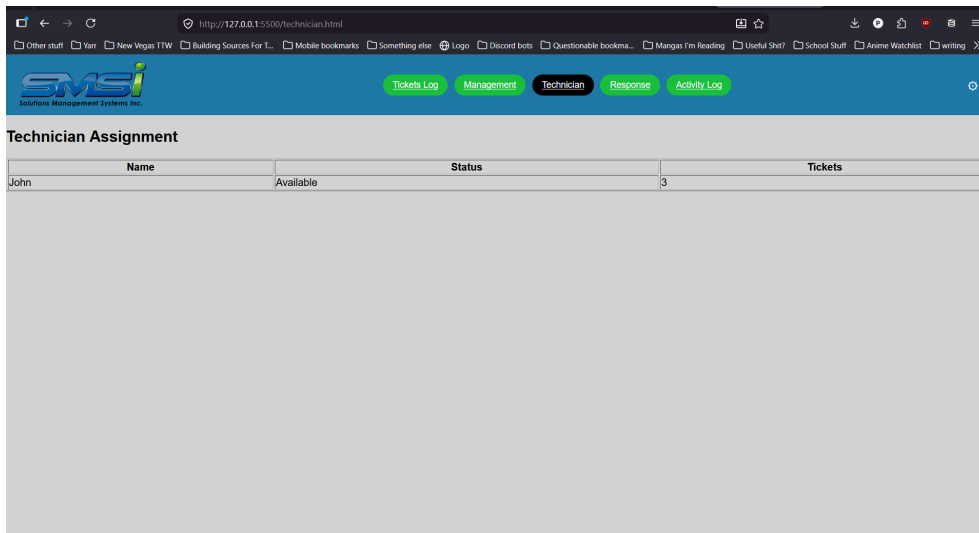
Dashboard:



Admin panel:



Data management page:



4. User Roles and Permissions

Administrator

- Full system access
- Manage users and clients
- Assign technicians
- View reports and logs

Staff / Technician

- View assigned work orders
- Update status of requests
- Add comments and activity logs

Regular User / Client

- Submit service requests
- View status of their requests

- Communicate through comments

5. Security Design

A. User Authentication

- Users must log in using email and password
- Sessions are created after successful login

B. Password Protection

- Passwords are encrypted before storing in the database
- Strong password rules are enforced

C. Access Control

- Role-Based Access Control (RBAC) is implemented
- Users can only access features based on their role
- Unauthorized access is restricted